WaveLKCell: Multi-Level Wavelet Enhancement for Large-Kernel Cell Nuclei Instance Segmentation

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***Abstract*—Accurate cell nuclei instance segmentation in hematoxylin and eosin (H&E) stained tissue images is essential for computational pathology. Recent large-kernel convolutional methods such as LKCell achieve state-of-the-art performance with efficient computation by expanding the receptive field through large convolution kernels. However, large kernels may blur fine high-frequency details critical for delineating nuclear boundaries and capturing chromatin texture. In this paper, we propose WaveLKCell, an architecture that integrates multi-level wavelet enhancement into the large-kernel segmentation framework to address this frequency limitation. Building upon the frequency-specific processing modules introduced in WaveConvX—including Adaptive Power Gabor Convolution (APGConv) for high-frequency details, self-attention for mid-high frequencies, channel attention for mid-low frequencies, and depthwise convolution for low frequencies—we design a U-shaped encoder-decoder that positions wavelet enhancement at the bottleneck between the large-kernel encoder and decoder. This architectural choice enables complementary frequency processing: large kernels capture broad spatial context while wavelet decomposition recovers high-frequency details. Experiments on the PanNuke benchmark demonstrate that WaveLKCell achieves competitive performance, achieving an mPQ of [X.XXXX] and a bPQ of [X.XXXX], validating the benefit of integrating wavelet-based frequency decomposition into large-kernel nuclei segmentation.**

***Index Terms*—nuclei instance segmentation, wavelet transform, large convolution kernel, computational pathology, frequency decomposition**

I. Introduction

Cell nuclei instance segmentation in hematoxylin and eosin (H&E) stained tissue images is a fundamental task in computational pathology, enabling quantitative analysis of tumor cell density, nucleus-to-cytoplasm ratio, and nuclear morphology for cancer grading and prognosis [?]. With the rapid development of deep learning, automated nuclei segmentation methods have emerged as powerful tools to reduce the burden on healthcare systems.

Despite significant progress, achieving high-performance and efficient nuclei segmentation remains challenging due to uneven staining, cell overlap, and clustered morphology. Previous convolutional neural network (CNN)-based methods [?], [?], [?] employ U-shaped architectures with small convolution kernels (typically 3×3) that limit the receptive field. While Vision Transformer (ViT)-based methods [?], [?] introduce global receptive fields, they incur substantial computational costs that hinder clinical deployment.

LKCell [?] addresses this trade-off by introducing large convolution kernels into nuclei segmentation for the first time, achieving state-of-the-art results with only 21.6% of the FLOPs compared to the previous leading method. However, large-kernel convolutions inherently apply spatial averaging over their receptive field, which can attenuate high-frequency information such as nuclear boundaries and fine chromatin patterns. These high-frequency details are precisely the cues needed for accurate instance separation and type classification.

To address this limitation, we draw inspiration from wavelet-based image analysis, which provides a principled framework for decomposing signals into frequency sub-bands. WaveConvX [?] recently demonstrated that multi-level wavelet enhancement with frequency-specific processing significantly improves histopathology image classification. This motivates the question: *can wavelet-based frequency decomposition complement large-kernel convolutions for nuclei segmentation?*

In this paper, we propose WaveLKCell, an architecture that integrates multi-level wavelet enhancement into the large-kernel nuclei segmentation framework. Our key insight is that wavelet decomposition and large-kernel convolutions operate on complementary frequency domains—large kernels capture broad spatial context in the low-frequency regime, while wavelet processing explicitly recovers and enhances the high-frequency details that large kernels tend to blur. Specifically, our contributions are:

- We design WaveLKCell, a U-shaped encoder-decoder architecture that integrates the multi-level wavelet enhancement modules of WaveConvX [?] into the LKCell [?] large-kernel segmentation framework, positioning wavelet frequency processing at the bottleneck to complement the large-kernel encoder's spatial feature extraction.
- We demonstrate that the combination of large-kernel convolutions and wavelet frequency decomposition is complementary for nuclei segmentation, where large kernels capture broad morphological context while wavelet processing recovers fine nuclear boundary and chromatin texture details.
- Experiments on the PanNuke dataset demonstrate competitive performance against state-of-the-art methods including CellPose-SAM, StarDist, LSP-DETR, and LKCell.

II. Related Work

A. Nuclei Instance Segmentation

Traditional image processing techniques for nuclei segmentation rely on hand-crafted features and domain knowledge.

The watershed algorithm [?] separates clustered nuclei using marker-controlled flooding, while approaches based on nuclear geometry require careful parameter tuning. These methods are inherently limited in representational capacity.

Deep learning has become the dominant paradigm. U-Net [?] established the encoder-decoder architecture with skip connections that has become the standard for medical image segmentation. HoVerNet [?] introduced the horizontal-vertical (HV) map representation alongside nuclear pixel (NP) and nuclear type (NT) predictions, using three separate decoder branches. CPP-Net [?] added parallel convolution layers for post-processing. StarDist [?] represents nuclei as star-convex polygons for instance separation. CellPose [?] uses gradient fields for cell boundary detection, and recent CellPose-SAM variants leverage foundation model features. LSP-DETR [?] adapts detection transformers for nuclei segmentation.

LKCell [?] demonstrated that large convolution kernels, previously successful in natural image analysis [?], [?], can achieve state-of-the-art nuclei segmentation with significantly reduced computation. However, large kernels may not optimally preserve high-frequency boundary details.

B. Wavelet Transforms in Medical Imaging

Wavelet transforms decompose signals into localized frequency sub-bands, enabling multi-resolution analysis. In medical imaging, wavelets have been used for denoising, feature extraction, and texture analysis. Recent works have integrated wavelets into deep learning pipelines. WaveMix [?] uses wavelet-based token mixing for image classification. WaveConvX [?] proposed multi-level wavelet enhancement within ConvNeXt for histopathology classification, demonstrating that frequency-specific processing improves feature representation.

Our work extends this idea from classification to instance segmentation, and specifically targets the complementary relationship between large-kernel convolutions and wavelet frequency decomposition.

III. Methodology

A. Overview

As illustrated in fig. 1, WaveLKCell consists of four main components: (1) a large-kernel encoder based on the UniRepLKNet architecture [?], (2) a multi-level wavelet enhancement module that processes the deepest encoder features, (3) a large-kernel decoder with skip connections, and (4) segmentation and classification heads.

Given an input H&E image $\mathbf{X} \in \mathbb{R}^{3\times H\times W}$, the encoder extracts multi-scale features $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ at three stages with dimensions $\{96, 192, 384\}$ and spatial resolutions $\{H/4, H/8, H/16\}$. The deepest features $\mathbf{F}_3$ are enhanced by the wavelet module to produce $\tilde{\mathbf{F}}_3$, which is then downsampled to $\mathbf{F}_4 \in \mathbb{R}^{768\times H/32\times W/32}$ for the decoder bottleneck.

The decoder progressively upsamples features through three large-kernel decoder blocks with skip connections from $\mathbf{F}_3$, $\mathbf{F}_2$, and $\mathbf{F}_1$, followed by two additional upsampling stages that incorporate early input features. Three segmentation heads predict the nuclear pixel (NP) binary map, horizontal-vertical (HV) map, and nuclear type (NT) map.

B. Large-Kernel Encoder

Following LKCell [?], we adopt the UniRepLKNet-S [?] architecture as our encoder. The encoder consists of three stages with depths $(3, 3, 27)$ and channel dimensions $(96, 192, 384)$. Each stage contains UniRepLKNet blocks with large depthwise convolution kernels.

Each UniRepLKNet block applies a large depthwise convolution (with kernel sizes of 13 or 3, following an alternating pattern), batch normalization, a Squeeze-and-Excitation (SE) block for channel attention, and a feed-forward network (FFN) with GRN (Global Response Normalization). The DilatedReparamBlock decomposes large kernels into multiple smaller dilated convolutions during training, which are reparameterized into a single convolution at inference for efficiency.

The encoder stem downsamples the input by $4\times$ through two consecutive strided convolutions. Each subsequent stage transition uses a strided convolution for $2\times$ downsampling. When available, we initialize the encoder with UniRepLKNet-S weights pre-trained on ImageNet.

C. Multi-Level Wavelet Enhancement

To address the high-frequency loss introduced by large kernels, we integrate the multi-level wavelet enhancement module of Zhang et al. [?] at the bottleneck of our architecture. This module decomposes the deepest encoder features $\mathbf{F}_3 \in \mathbb{R}^{C\times H'\times W'}$ $(C = 384)$ into frequency sub-bands via two-level Haar DWT, applies frequency-specific processing to each sub-band, and reconstructs enhanced features via IDWT.

1) Wavelet Decomposition: A two-level Haar DWT is applied to $\mathbf{F}_3$, yielding seven sub-bands:

$$\{\mathrm{LL}_2, \mathrm{LH}_2, \mathrm{HL}_2, \mathrm{HH}_2, \mathrm{LH}_1, \mathrm{HL}_1, \mathrm{HH}_1\} \tag{1}$$

at resolutions $H'/4 \times W'/4$ and $H'/2 \times W'/2$, where LL contains low-frequency approximations and LH, HL, HH capture horizontal, vertical, and diagonal detail coefficients, respectively. The DWT is implemented via strided convolutions with fixed Haar kernels [?].

2) Frequency-Specific Processing: As proposed in WaveConvX [?], each frequency group is processed by a dedicated module (fig. 2). Rather than detailing each processor here (see [?] for full descriptions), we summarize the processing strategy and highlight adaptations made for our segmentation task:

High Frequency ($\mathbf{HH}_1$) is processed by Adaptive Power Gabor Convolution (APGConv) [?], which applies learnable Gabor filters with an adaptive power exponent α_k. This exponent controls the nonlinearity of the filter response: $\alpha_k < 1$ amplifies subtle texture variations, while $\alpha_k > 1$ emphasizes strong edges, making it particularly suited for nuclear boundary detection.

Mid-High Frequency ($\mathbf{LH}_1$, $\mathbf{HL}_1$) is processed by multi-head self-attention (4 heads) to model spatial dependencies

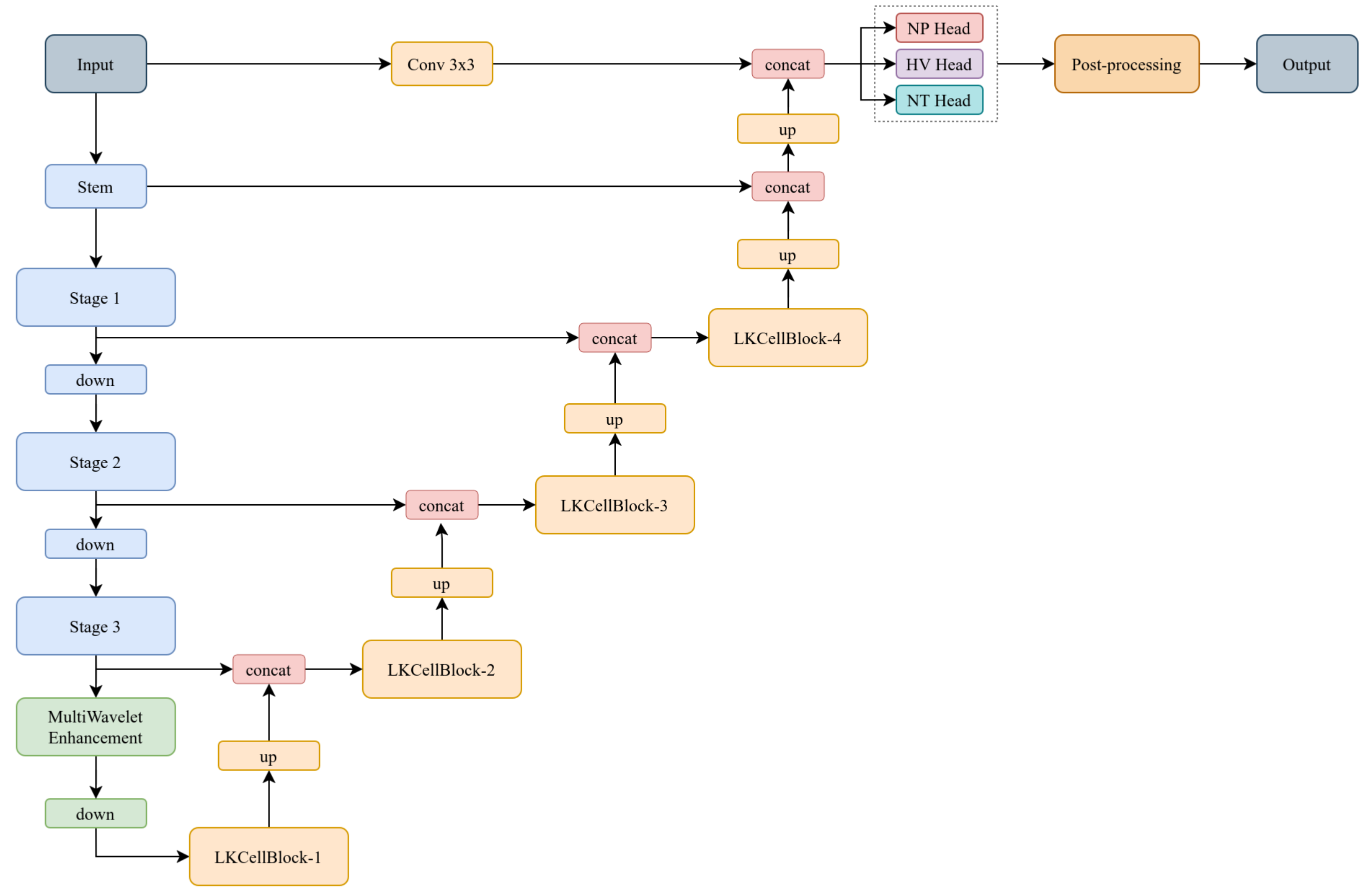

Fig. 1: Overview of the WaveLKCell architecture in a U-shaped encoder-decoder layout. The large-kernel encoder (UniRepLKNet-S) extracts multi-scale features $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ at progressively lower resolutions. The Multi-Level Wavelet Enhancement module processes $\mathbf{F}_3$ through two-level Haar DWT/IDWT with frequency-specific processors (fig. 2), producing the bottleneck $\mathbf{F}_4$. The decoder upsamples through three large-kernel blocks with skip connections, followed by two extra upsampling stages. Three heads predict NP, HV, and NT maps.

in transitional edge structures [?]. A key difference from WaveConvX is that we process LH_1 and HL_1 independently with shared weights rather than averaging them, preserving the directional information required by the subsequent IDWT reconstruction.

Mid Frequency ($\mathbf{HH}_2$) is processed by depthwise separable convolution for local feature refinement.

Mid-Low Frequency ($\mathbf{LH}_2$, $\mathbf{HL}_2$) is processed by squeeze-and-excitation channel attention [?], followed by a 3×3 convolution, to recalibrate channel-wise feature responses.

Low Frequency ($\mathbf{LL}_2$) is processed by a standard 3×3 convolution with batch normalization and GELU activation.

3) Reconstruction: The enhanced sub-bands are reconstructed via two-level IDWT:

$$\mathrm{LL}_1' = \mathrm{IDWT}(\mathrm{LL}_2', \mathrm{LH}_2', \mathrm{HL}_2', \mathrm{HH}_2'), \quad \tilde{\mathbf{F}}_3 = \mathrm{IDWT}(\mathrm{LL}_1', \mathrm{LH}_1', \mathrm{HL}_1', \mathrm{HH}_1') \tag{2}$$

A residual connection preserves the original features:

$$\hat{\mathbf{F}}_3 = \mathbf{F}_3 + \tilde{\mathbf{F}}_3 \tag{3}$$

The wavelet-enhanced features $\hat{\mathbf{F}}_3$ are then downsampled by a strided convolution with layer normalization to produce the bottleneck features $\mathbf{F}_4 \in \mathbb{R}^{768 \times H/32 \times W/32}$ for the decoder.

D. Large-Kernel Decoder

The decoder follows the LKCell [?] design with ReparamLargeKernelConv blocks. Each decoder block consists of:

1) A large depthwise convolution (kernel sizes 13, 27, 29 for the three blocks) combined with a small depthwise convolution (5×5) via the ReparamLargeKernelConv module, with batch normalization, ReLU activation, and a 1×1 pointwise convolution.
2) Bilinear 2× upsampling.
3) Concatenation with the corresponding encoder skip connection features, followed by a 1×1 convolution for channel alignment.
4) A ConvFFN module with a 4× expansion ratio and drop path regularization.

After the three large-kernel blocks, two additional upsampling stages incorporate early input features (at 1× and 1/2×

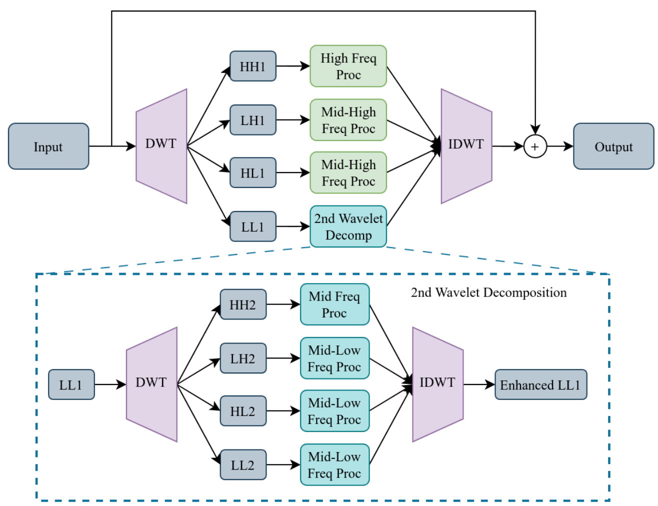

Fig. 2: Detailed view of the Multi-Level Wavelet Enhancement module. The input $\mathbf{F}_3$ is decomposed by two-level Haar DWT. Each frequency sub-band is processed by a specialized module: APGConv for HH_1 (high-frequency boundaries), self-attention for LH_1/HL_1 (mid-high), channel attention for LH_2/HL_2 (mid-low), depthwise conv for HH_2, and conv for LL_2 (lowest). Enhanced bands are reconstructed via IDWT with a residual connection.

resolution) through strided convolutions from the input stem, further refining spatial details.

The decoder output is fed to three segmentation heads, each consisting of batch normalization, GELU, 1×1 convolution, batch normalization, GELU, and a final 1×1 convolution:

- **NP Head**: Predicts a 2-channel binary map (background vs. nucleus).
- **HV Head**: Predicts a 2-channel horizontal-vertical regression map.
- **NT Head**: Predicts an N_c-channel nuclear type map (N_c nucleus classes).

E. Training Objectives

The total training loss combines three terms:

$$\mathcal{L} = \lambda_{\text{np}}\mathcal{L}_{\text{NP}} + \lambda_{\text{hv}}\mathcal{L}_{\text{HV}} + \lambda_{\text{type}}\mathcal{L}_{\text{type}} \tag{4}$$

where $\mathcal{L}_{\text{NP}}$ and $\mathcal{L}_{\text{type}}$ are cross-entropy losses for binary and type prediction, and $\mathcal{L}_{\text{HV}}$ is an MSE loss for the HV regression map. We set $\lambda_{\text{np}} = \lambda_{\text{hv}} = \lambda_{\text{type}} = 1.0$.

F. Post-Processing

Following HoVerNet [?], we convert the predicted maps to instance segmentation using marker-controlled watershed. The HV map provides gradient information for watershed segmentation, while the NP binary map defines the foreground mask. Nuclear types are assigned via majority voting within each instance.

TABLE I: Comparison with state-of-the-art methods on the PanNuke dataset. Best results are in **bold**. † indicates numbers from original papers.

Method	mPQ↑	bPQ↑
StarDist [?]	[X.XXXX]†	[X.XXXX]†
CellPose-SAM [?]	[X.XXXX]†	[X.XXXX]†
LSP-DETR [?]	[X.XXXX]†	[X.XXXX]†
LKCell [?]	0.5080†	0.6847†
WaveLKCell (Ours)	**[X.XXXX]**	**[X.XXXX]**

TABLE II: Ablation study on the wavelet enhancement module.

Configuration	mPQ↑	bPQ↑
LKCell baseline (no wavelet)	[X.XXXX]	[X.XXXX]
+ Wavelet (all conv processors)	[X.XXXX]	[X.XXXX]
+ Wavelet (specialized processors)	[X.XXXX]	[X.XXXX]

IV. Experiments

A. Dataset and Metrics

PanNuke [?] is a large-scale nuclei instance segmentation dataset containing 189,744 semi-automatically annotated nuclear patches from 19 tissue types. Each nucleus is labeled with one of 5 categories: Neoplastic, Inflammatory, Connective, Dead, and Non-neoplastic Epithelial. We use the standard fold split with 3-fold cross-validation.

We evaluate using the **Panoptic Quality (PQ)** metric, which combines segmentation quality and detection accuracy:

$$\text{PQ} = \frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|} \times \text{SQ} \tag{5}$$

where SQ (Segmentation Quality) is the mean IoU of true positive pairs. We report **mPQ** (mean PQ across tissue types), **bPQ** (binary PQ ignoring types), and per-class PQ.

B. Implementation Details

We train WaveLKCell using AdamW optimizer with a cosine annealing learning rate schedule. The backbone encoder uses a $10\times$ smaller learning rate. Input images are padded or cropped to 256×256 pixels. The encoder is initialized with UniRepLKNet-S weights pre-trained on ImageNet. All experiments are conducted using PyTorch Lightning on NVIDIA GPUs.

C. Comparison with State-of-the-Art

table I presents the comparison with state-of-the-art methods. WaveLKCell demonstrates competitive performance against strong baselines. The wavelet enhancement module provides improvements over the base LKCell architecture, particularly in [TODO: specify which metrics/tissue types benefit most].

D. Ablation Studies

We conduct ablation studies to validate each design choice.

1) Effect of Wavelet Enhancement: table II shows that adding the wavelet module improves segmentation quality over the LKCell baseline. Using specialized frequency-specific processors (APGConv, self-attention, channel attention) yields further gains over replacing all processors with standard convolutions, confirming that tailored frequency processing is beneficial.

V. Conclusion

We presented WaveLKCell, an architecture that integrates multi-level wavelet enhancement into a large-kernel nuclei instance segmentation framework. By positioning the frequency-specific processors of WaveConvX [**?**] at the bottleneck of the LKCell [**?**] encoder-decoder, our architecture leverages the complementary strengths of large-kernel convolutions for broad spatial context and wavelet decomposition for high-frequency detail recovery. Experiments on the PanNuke benchmark demonstrate competitive performance against state-of-the-art methods, validating the benefit of combining these two approaches for nuclei segmentation.

In future work, we plan to explore learnable wavelet bases, extend the wavelet enhancement to multiple decoder stages, and evaluate on additional pathology datasets including MoNuSeg and CoNIC.